

H33_01

The equation $y^3 + y^2 + xy = x^4 + 1$ has no integer solutions

Size $h = 33$, length $l = 11$, identifier E0258 (with y replaced by $-y$)

Theorem 1. *The equation*

$$y^3 + y^2 + xy = x^4 + 1 \tag{1}$$

has no integer solutions.

The proof uses a polynomial identity of the form $HT = U^2 + V^2$ on the curve (1), and the fact that a prime p with $p \equiv 3 \pmod{4}$ divides $U^2 + V^2$ only to an even power. Apart from polynomial identities, which can be checked by expanding both sides, and finite checks of congruences, the proof uses only Fermat's little theorem.

1 The norm form $a^2 + b^2$

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N .

Lemma 2. *Let p be a prime with $p \equiv 3 \pmod{4}$, and let $a, b \in \mathbb{Z}$. If $p \mid a^2 + b^2$, then $p \mid a$ and $p \mid b$.*

Proof. Suppose that $p \nmid b$. Choose $c \in \mathbb{Z}$ with $bc \equiv 1 \pmod{p}$ and put $r = ac$. Then $r^2 \equiv -(bc)^2 \equiv -1 \pmod{p}$, so $p \nmid r$, and Fermat's little theorem gives

$$1 \equiv r^{p-1} = (r^2)^{(p-1)/2} \equiv (-1)^{(p-1)/2} = -1 \pmod{p},$$

because $(p-1)/2$ is odd. Then $p \mid 2$, which is impossible. Hence $p \mid b$, and then $p \mid a^2$, so $p \mid a$. $\square$

Lemma 3. *Let p be a prime with $p \equiv 3 \pmod{4}$, and let $a, b \in \mathbb{Z}$ be not both zero. Then $v_p(a^2 + b^2)$ is even.*

Proof. Let p^k be the largest power of p dividing both a and b , and write $a = p^k a'$ and $b = p^k b'$. Then p does not divide both a' and b' , so $p \nmid a'^2 + b'^2$ by Lemma 2. Since $a^2 + b^2 = p^{2k}(a'^2 + b'^2)$, we get $v_p(a^2 + b^2) = 2k$. $\square$

2 The auxiliary polynomials

Put $F(x, y) = y^3 + y^2 + xy - x^4 - 1$, so that (1) is the equation $F(x, y) = 0$. Define

$$\begin{aligned} H &= x^2 + 2x + 2y, \\ T &= 4y^2 + (4 - 2x^2 - 4x)y + (x^2 + 2x)^2 - 2x^2, \\ U &= x^3 + 3x^2 + 2, \quad V = 2 - 3x^2, \\ G &= 8, \end{aligned}$$

and

$$A = 9(x - 6), \quad B = 3x^2 - 9x - 52.$$

Expanding both sides, one checks the identities

$$HT - U^2 - V^2 = GF, \quad (2)$$

$$H = (x+1)^2 + 2y - 1, \quad (3)$$

$$AU + BV = -212 = -2^2 \cdot 53. \quad (4)$$

3 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1). Then $F(x, y) = 0$, and (2) gives

$$HT = U^2 + V^2, \quad (5)$$

where H, T, U and V now denote the values of these polynomials at (x, y) .

Step 1: congruences. Checking the 16 residue pairs modulo 4, we find that the only solutions of (1) modulo 4 are $(x, y) \equiv (1, 2)$ and $(x, y) \equiv (3, 2)$. In both cases $H \equiv 3 \pmod{4}$. In particular, x is odd and y is even.

Step 2: signs. If $y = 0$, then (1) gives $x^4 + 1 = 0$, which is impossible. Suppose that $y < 0$. Since y is even, $z = -y \geq 2$, and (1) becomes $x^4 + xz + z^3 - z^2 + 1 = 0$. But $x^4 \geq x^2$ for every integer x , and $z^3 - z^2 = z^2(z - 1) \geq z^2$, so the left-hand side is at least $x^2 + xz + z^2 + 1 = (x + z/2)^2 + 3z^2/4 + 1 > 0$, a contradiction. Hence $y \geq 2$, and (3) gives $H \geq 3 > 0$.

Step 3: primes with $p \equiv 3 \pmod{4}$. Let p be a prime with $p \equiv 3 \pmod{4}$. We show that $v_p(H)$ is even. By (4), U and V are not both zero, so (5) and Lemma 3 show that $v_p(HT) = v_p(U^2 + V^2)$ is even. Suppose that p divides both H and T . Then $p \mid U^2 + V^2$ by (5), so $p \mid U$ and $p \mid V$ by Lemma 2, and $p \mid 2^2 \cdot 53$ by (4). But $2 \not\equiv 3 \pmod{4}$ and $53 \equiv 1 \pmod{4}$, a contradiction. Hence p does not divide both H and T . If $p \nmid T$, then $v_p(H) = v_p(HT)$ is even, and if $p \nmid H$, then $v_p(H) = 0$.

Step 4: contradiction. By Steps 1 and 2, H is a positive odd integer. Every prime factor p of H with $p \not\equiv 3 \pmod{4}$ satisfies $p \equiv 1 \pmod{4}$, and every prime factor with $p \equiv 3 \pmod{4}$ occurs with an even exponent by Step 3. Since the residue 1 modulo 4 is closed under multiplication and contains the squares of all odd integers, $H \equiv 1 \pmod{4}$. This contradicts Step 1, and the theorem is proved. $\square$